ResumeLens

Deterministic résumé screening with regular expressions, transducers, automata and a DSL

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Same résumé, same verdict: a fully formal pipeline with no LLM, so nothing is guessed.

Same skill, different spellings.
`JS → JAVASCRIPT` `sklearn → SCIKIT_LEARN`

1 · EXTRACTION

Regular expressions

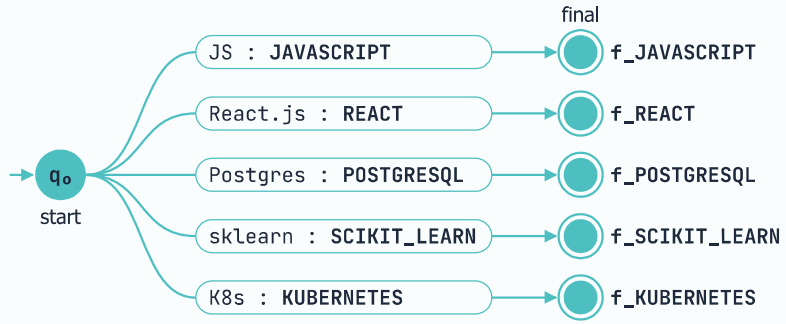
`(?P<skill>…)` `(?P<user>…)` `(?P<domain>…)` `(?P<years>…)` `(?P<role>…)` `(?P<company>…)` `(?P<degree>…)` `(?P<location>…)` `(?P<text>…)`

Wednesday Addams
`JS · React.js · NodeJS · Postgres · Git`

Mary Jane Watson
`Python · Pandas · NumPy · Scikit-learn · TensorFlow · SQL · Git`

Also extracted: location, current role, summary, job bullets

Fig. 1 · FST: spellings → canonical skill



2 · NORMALIZATION

Finite-State Transducer (FST)

$M = (Q, \Sigma, \Gamma, \delta, \omega, q_0, F)$

Full Stack order:
`Language → Frontend → Backend → Database → API → VCS (version control)`

`JS, React.js, NodeJS, Postgres, Git`
⟹
`JAVASCRIPT, REACT, NODE_JS, POSTGRESQL, GIT`

3 · CLASSIFICATION

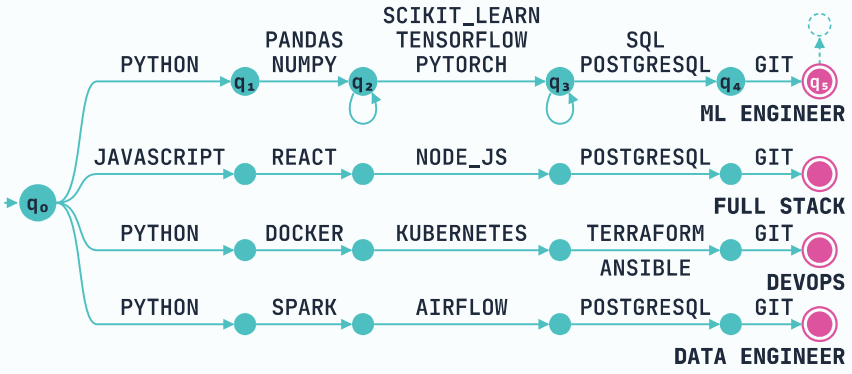
Finite automata

$M = (Q, \Sigma, \delta, q_0, F)$

Accepted only if the skill sequence reaches a final state.

Full Stack	**ML**
DevOps	**Data Engineer**

Fig. 2 · Profile automata



4 · SPECIFICATION

Domain-Specific Language (DSL) · textX grammar

$G = (V, \Sigma, S, P)$

```
CandidateReport: 'candidate' '{'
  personal=PersonalInfo
  (summary=SummarySection)?
  education=EducationSection
  experience=ExperienceSection
  skills=SkillSection
  evaluation=EvaluationSection '}' ;
```

Fig. 3 · Verdict matrix

	Full Stack	ML	DevOps	Data
Full Stack CV	●			
ML CV		●		
DevOps CV			●	
Data CV				●
Corrupt CV				

Evaluated on 5 synthetic résumés (4 valid, 1 corrupt).

Results

5/5 CVs with the expected verdict

1,864 automated tests passing

4 profiles verified

Conclusion

Formal models make every decision traceable: each verdict is a path through an automaton.

Limitations & next steps

- Closed vocabulary: unknown spellings are flagged, not guessed.
- Next: validate on real résumés and extend the lexicon.

WA Wednesday Addams

Location: Nevermore Academy, Jericho
Current role: Student and independent investigator

Profile Summary
3 years of experience developing web applications.

Experience
Full Stack Developer · Raven Labs · 2023 - 2026
- Built a React.js dashboard backed by a NodeJS REST API.

`JAVASCRIPT` `REACT` `NODE_JS` `POSTGRESQL` `GIT`

Full Stack Developer – ACCEPTED

✓ VERIFIED VIA DETERMINISTIC AUTOMATA

Built with Python, pyformlang, textX and Streamlit.